Creative Elements and Novelty

Creative / novel components

1. Structure- rst framing of genre remastering instead of shallow timbre swapping.
2. A staged pipeline with explicit deconstruction, calibration, reconstruction, and coherence stages.
3. Direct-output codec translation plus MERT-based conditioning.
4. Pre x-lock coherence constraints inside di usion sampling.

Why these count

They are documented engineering responses to observed bottlenecks in the repo, not generic o - the-shelf invocations.

How to present this

Emphasize that the novelty is not one isolated block. It is the combination of representation learning, stronger conditional translation, and long-form sampling controls.

Creative-elements angle for CMPUT 414

The project contributes new engineering structure and tuning decisions rather than merely re-running a known model with default settings.

Limitations and Lab 5

Current limitations

1. Dataset-source leakage is still a genre-label risk.
2. Di usion quality currently lags codec transfer on short-form reliability.
3. Long-form audio still accumulates warble and static under stronger edits.

Lab 5 next step

Run a blind listening study against simpler lter-style baselines and report both subjective genre authenticity and objective realism proxies.

What the final study should answer

Can listeners identify the target genre? Can they still recognize the source melody? Do they prefer our remaster to a simpler baseline?

Why this is the right ending

At this point the key open question is no longer whether the architecture runs. It is whether people actually hear the intended e ect.

Conclusion

Lecture takeaway

Serious music style transfer is not one trick. It needs a representation story, a conditioning story, a synthesis story, and a coherence story.

One-sentence summary

We started from the general problem of meaningful genre transfer and built a repo-backed system that implements one defensible answer end-to-end.

Case-study takeaway

DGGR already clears its short-form technical gates and extends into measurable long-form coherence. The remaining gap is mainly perceptual re nement and formal human evaluation.

Discussion prompts

Codec editing, di usion, or hybrid?

How should music evaluation balance taste with repeatability?

Backup: Exact Codec Branch Dimensions

Component	Implementation detail
Input latents	EnCodec quantized embeddings, approximately $[B; 128; T]$
Condition vector	$z_c \in \mathbb{R}^{128}$, target style embedding $z_s \in \mathbb{R}^{128}$, noise $\in \mathbb{R}^{32}$
Hidden width	256 channels
Residual stack	10 FiLM-conditioned residual Conv1D blocks
Dilations	$[1; 2; 4; 8; 1; 2; 4; 8; 1; 2]$
Normalization	GroupNorm with 8 groups

Backup: Exact Di□usion Branch Dimensions

Component	Implementation detail
Input tensor	$[B; 15; 80; 432]$ = 1 noisy mel +12 chroma +1 onset +1 beat channel
Output tensor	$[B; 1; 80; 432]$ velocity/noise prediction
UNet channels	$[64; 128; 256; 256]$
Residual blocks	2 per resolution level
Attention	lower resolutions in the current code path
Condition split	time and content via FiLM, style via dedicated AdaIN

Backup: Repo Evidence Used for This Lecture

Core repo-backed evidence

Lab 1 explanations and audits in docs/explanation/lab1_deconstruction_encoder.md and saves/lab1_run_combo_af_gate_exit_v2/.
Lab 2 calibration artifacts in saves/lab2_calibration/lab2_20260211_015118_lda_cleanup_v2/.
Codec branch code in dggr/lab3_codec_models.py; best metrics in saves2/lab3_codec_transfer/run1055/.
Di usion branch code in dggr/lab3_diffusion_model.py; training history in saves2/lab3_diffusion/run_d002/.
Long-form coherence code in lab 3/run_lab4_longform_coherence.py; diagnostics in saves2/lab4_longform_coherence/fullsong_test/.

Visual and media assets

Charts and spectrograms were generated locally from repo artifacts using

Backup: Lecture References

Dai et al. (2018), Aucouturier and Pachet (2003), Ghosal and Kolekar (2018), Hung et al. (2019), Yang et al. (2019), Roberts et al.